

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is broadly compatible with deployment design. Discrete-label NER outputs with IOB2 [\[Q52\]](#) and continuous STS scores [\[Q54\]](#) match the deployment's multi-candidate confidence-score architecture. Multi-label classification format (one-hot encoding in MORFITT-D4, MORFITT-D21, DEFT2021-D4) supports the multi-candidate output design. Statistical significance reporting via Student's t-test [\[Q51\]](#) and four-run averaging support reliability claims. However, the benchmark does not evaluate confidence calibration, uncertainty quantification, or threshold selection for human-review escalation — operationally critical parameters for the deployment's borderline-case handling. The deployment's threshold for mandatory escalation is unverified in deployment documentation [region YAML threshold_documentation NEEDS VERIFICATION]. DEFT2020's retention of individual annotator scores (DEFT2020-D12, DEFT2020-D13) provides a partial basis for calibration analysis. Severe class imbalances in CAS (74% neutral, CAS-D16), ESSAI (76.8% neutral), and PxCorpus (replace=3, PXCORPUS-D21) limit evaluation of safety-critical minority classes — highly relevant for compliance flagging. Silver-standard POS labels (CAS-D14, ESSAI-D12) propagate label noise into output evaluation.

ID	Evidence
Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q51	"All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test." (p.5)
MORFITT-D4	Les échanges internationaux sont responsables de la résurgence de nombreuses maladies affectant le bétail. — Livestock disease / international trade — multi-label veterinary context
MORFITT-D21	Bloquer le complément, notamment l'axe C5a-C5aR1, par des thérapies spécifiques représente un espoir thérapeutique dans les formes les plus sévères de la maladie. — COVID-19 complement pathway — three-label multi-label assignment
DEFT2021-D4	(six specialty labels for complex multi-system hepatotoxicity case) — Demonstrates multi-label output consistent with deployment's multi-candidate design
DEFT2020-D12	scores=[5.0, 2.0, 0.0, 3.0, 1.0] — High inter-annotator disagreement; signals annotation uncertainty
DEFT2020-D13	scores=[2.0, 4.5, 0.0, 1.0, 0.0] — Extreme annotator divergence on low-similarity pair
CAS-D16	negation_speculation: 6, negation: 100, neutral: 368, speculation: 26 — Extreme neutral class dominance — minority safety-critical classes underrepresented
PXCORPUS-D21	medical_prescription: 447, negate: 11, replace: 3, none: 39 — Severe class imbalance; `replace` (correction) class underrepresented
CAS-D14	tokens: ['le', 'scanner', 'réalisé', ...] pos_tags: [12, 24, 25, ...] — Automatic POS assignment without full human adjudication — silver-standard risk
ESSAI-D12	Le MEDI9197 est injecté en intra-tumoral tous les 28j (4 semaines). — Drug identifier MEDI9197 tagged as NAM (tag 7) by automatic TreeTagger — silver standard

Strengths

- Discrete NER labels and continuous STS scores match deployment's multi-candidate confidence-score output design [Q52, Q54]
- Four-run averaging with Student's t-test significance testing supports reliability [Q51]
- DEFT2020 retains individual annotator scores enabling uncertainty/calibration analysis (DEFT2020-D12, DEFT2020-D13)
- Tokenizer-agnostic SeqEval evaluation [Q53] supports robust comparison across model tokenizers
- Multi-label one-hot output compatible with multi-candidate design (MORFITT-D4, DEFT2021-D4)

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Q53	"It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process." (p.6)
MORFITT-D4	Les échanges internationaux sont responsables de la résurgence de nombreuses maladies affectant le bétail. — Livestock disease / international trade — multi-label veterinary context
DEFT2021-D4	(six specialty labels for complex multi-system hepatotoxicity case) — Demonstrates multi-label output consistent with deployment's multi-candidate design

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — output modality (discrete labels, continuous scores) matches the deployment's multi-candidate confidence-score and human-review-flagging design.

OF-2: N/A — deployment is text-only; no speech output requirement.

OF-3: N/A for primary user population (regulatory affairs specialists, high literacy assumed). The deployment is professional-internal, not patient-facing.

OF-4: Moderate gap: benchmark does not evaluate confidence calibration, uncertainty quantification, or human-review escalation thresholds [Q82]. Severe class imbalances in CAS (CAS-D16), ESSAI, and PxCorpus (PXCORPUS-D21) limit evaluation of minority safety-critical classes — directly relevant to compliance flagging.

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Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)

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DEFT2021-D4	(six specialty labels for complex multi-system hepatotoxicity case) — Demonstrates multi-label output consistent with deployment's multi-candidate design
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
WEB-6	Variation types IA/IB/II directly map to threshold-based escalation requirements but are not evaluated in benchmark (https://alhena-consult.com/how-to-market-a-drug-in-france/)
CAS-D16	negation_speculation: 6, negation: 100, neutral: 368, speculation: 26 — Extreme neutral class dominance — minority safety-critical classes underrepresented
PXCORPUS-D 21	medical_prescription: 447, negate: 11, replace: 3, none: 39 — Severe class imbalance; `replace` (correction) class underrepresented
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Information Gaps

- Deployment threshold for mandatory human review escalation is not documented [region YAML threshold_documentation NEEDS VERIFICATION]
- Whether benchmark output formats specifically support per-class confidence scores or only top-1 predictions

Requires Expert Verification

- Calibration thresholds at which borderline STS scores should trigger mandatory human review under EMA/ANSM compliance workflows

Remediation

Gap 1: Benchmark does not evaluate confidence calibration, uncertainty quantification, or threshold selection for human-review escalation — operationally critical for the human-in-the-loop compliance workflow.

Recommendation: Use DEFT2020's retained individual annotator scores (DEFT2020-D12, DEFT2020-D13) and CLISTER's score distributions to perform calibration analysis on candidate models. Define and validate confidence thresholds for mandatory human review against regulatory affairs workflow requirements before deployment.

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